Reply to NVIDIA Customer Care (Aadya) — DGX Spark thermal case

Hi Aadya, thank you for the quick response. Answers to every item below, with measured telemetry. Summary up front: the unit hard powers off (instant, no shutdown sequence, nothing in logs) when platform thermal zones reach ~91°C, which occurs within ~45 seconds of sustained GPU load at under 100 W. The GPU die sensor stays 12–15°C BELOW the platform zones throughout, which points to degraded heat transfer (TIM/heatsink contact) rather than fan or ambient.

1. Does the system feel unusually hot, throttle, freeze, restart, or shut down? It hard shuts down (instant power-off, no OS shutdown sequence, system journal cut mid-line). Four occurrences: Aug 22 08:02, Aug 26 02:15, Aug 26 12:05, Aug 26 15:48 (all logged in journalctl boot history). Before shutdown the BMC appears to cap GPU power at ~60–95 W despite 95%+ utilization, with no kernel throttle-reason flag set.

2. GPU temperature at idle and under the workload - Idle (measured just now): GPU die 44°C at 11.7 W, ACPI platform zones 44–47°C. - Under load: GPU die 64–83°C — but ACPI platform zones (acpitz) climb from ~80°C to 89–91°C within 45 seconds at 92–97 W. The 12–15°C die-vs-platform gap persists throughout.

3. Screenshot / nvidia-smi output Attached: (a) controlled load test log — timestamps, per-zone temps, nvidia-smi power/temp/util every ~11 s until safety abort at 89.4°C; (b) a black-box telemetry capture (30-second fsync'd samples) of the actual Aug 26 15:48 shutdown: last recorded sample shows 96% util, 67 W, zones at 91.2°C, followed by instant power loss. [attach results/thermal_verify.log and the final section of results/blackbox.log; screenshots of nvidia-smi can be provided on request]

4. Workload running when the issue occurs PyTorch 2.13 (cu130) language-model training, native (no container), batch size 16–32, ~120M-parameter model. Also reproduced with a plain PyTorch matmul loop — the issue is load-dependent, not application-specific.

5. Time for the problem to appear Thermal runaway begins immediately under load: platform zones go 79.6°C → 89.4°C in 44 seconds at ≤97 W. Hard shutdowns occurred between ~2 minutes and ~4.3 hours into training runs (duty-cycling workloads last longer). Recovery is equally fast when load stops (85°C → 68°C within 10–20 s), consistent with a localized heat-transfer defect.

6. Ambient room temperature [OPERATOR: fill in, e.g. "~22°C, air-conditioned room"] — well within the 5–30°C range. Note: an RTX 5090 workstation in the same room sustains 483 W at 69°C GPU, so ambient is demonstrably not the cause.

7. Ventilation / vents unobstructed? [OPERATOR: confirm, e.g. "Yes — unit is on an open desk, vents clear, nothing stacked on or around it."]

8. Fan spinning / unusual noise? [OPERATOR: confirm what you hear under load.] No fan RPM sensor is exposed to the OS (no hwmon fan inputs), so we cannot verify in software. The very fast cool-down when load stops suggests airflow exists; the defect pattern matches heatsink/TIM contact rather than a stopped fan.

9. Supplied NVIDIA 240 W adapter in use? [OPERATOR: confirm "Yes, original NVIDIA adapter and cable."]

10. DGX OS, driver, firmware versions - DGX OS: 7.2.3 (build 2025-09-10, commit 833b4a7), Ubuntu 24.04.4 LTS - Kernel: 6.17.0-1031-nvidia (issue also reproduced on 6.17.0-1029-nvidia) - GPU driver: 580.173.02 (open kernel module) - UEFI/system firmware: 0x03000508; UEFI PK 2025; NVMe FW NXHB202Q - fwupdmgr reports Embedded Controller and UEFI at latest, no pending updates

11. All DGX Spark updates installed? Yes — fwupdmgr shows no available firmware updates; OS kernel is current (6.17.0-1031-nvidia, installed Aug 26). A handful of routine apt package updates are pending but none are firmware/driver/thermal related.

12. Connected devices Headless operation: no display, no external storage. Only internal WiFi/BT module on USB; gigabit LAN connected. Nothing else attached.

13. Opened, modified, or physically damaged? No. The unit has never been opened or modified and has no physical damage.

14. Screenshots of thermal warnings/errors There are none — that is the core symptom. The shutdown is instantaneous with no OS-level warning, message, or shutdown sequence; the system journal simply stops mid-line (timestamps above). Our own fsync'd telemetry recorder captured the final seconds (attached).

15. Troubleshooting already completed - Ruled out: memory pressure (sar shows ~20% RAM used at every event), disk (1.3 TB free), kernel version (reproduced on 1029 and 1031), firmware level (EC/UEFI current), OOM/panic (none logged), ambient (see #6). - Controlled reproduction with safety abort: documented 10°C/30 s runaway at <100 W with a persistent 12–15°C die-vs-platform gap. - Interim mitigation in place: a software governor pauses workloads above 85°C (SIGSTOP/SIGCONT), which prevents shutdowns but caps the machine at a fraction of its rated capability. - **Field Diagnostics**: dgx-spark-fieldiag is installed; `partnerdiag --field` fails at MODS driver insertion because Secure Boot is enabled ("Key was rejected by service") and this unit runs headless (no console available to complete a MokManager/Secure Boot change remotely). We can arrange one-time physical access to disable Secure Boot and run FieldDiag if required — please confirm whether you need it to proceed, given the telemetry above.

Serial number: [OPERATOR: output of `sudo dmidecode -s system-serial-number`]

We believe this matches the known GB10 thermal-defect pattern discussed in developer-forum threads 349647, 363370, 373266 and 380238 (the latter two resulting in RMAs). Requesting RMA evaluation.